DAYANANDA SAGAR UNIVERSITY
SCHOOL OF ENGINEERING
Department of Computer Science and Engineering | Academic Year 2026-2027

SELF-CALIBRATING MULTI-MODAL EDGE AI NODE FOR INDUSTRIAL MACHINE ANOMALY DETECTION

Major Project Phase-I (Zeroth Review)

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Problem Definition

→ Problem Statement

- In ***Small and Medium Enterprises (SME's)***, heavy machines like fans, pumps, and motors run continuously. When a machine part starts wearing out, workers rarely notice it until the machine breaks down completely. This causes huge irreversible repair costs and stops work for days.
- Existing commercial monitoring systems *cost per machine* and depend on continuous cloud connectivity — which is unreliable inside metal factory buildings with ***weak Wi-Fi/4G.***

→ Motivation & Need

- Unplanned downtime disrupts production schedules and causes ***direct financial loss*** to small manufacturers who cannot absorb the cost of enterprise-grade monitoring.
- Damaged, poorly maintained motors also consume ***15–30% more electricity*** than healthy ones, an ongoing hidden cost beyond the breakdown itself.

→ Project Relevance

- Addresses a real, underserved gap in affordable industrial monitoring for MSMEs, a segment largely excluded from existing IoT solutions.
- SIH Theme: ***Smart Industrial Automation*** / Industry 4.0 / Industrial IoT
- UN SDGs: ***SDG 7*** (Affordable & Clean Energy), ***SDG 9*** (Industry, Innovation & Infrastructure), ***SDG 12*** (Responsible Consumption & Production)

Social/Environmental/ Technical Impact

- **Social Impact:**

  • Protects factory operators from hazardous machine explosions, electrical fires, and snapped high-speed fan blades.
  • Democratizes Industry 4.0 safety, allowing small Indian factory owners to protect their workforce and equipment affordably.

- **Environmental Impact:**

  • Reduces industrial e-waste and heavy machinery scrap by detecting component wear early before complete motor destruction.
  • Cuts carbon emissions by eliminating 15%–30% excess electrical power drawn by unmaintained, wobbly motors.

- **Technical Impact:**

  • Advances TinyML system integration on low-cost microcontrollers by demonstrating real-time multi-modal sensor fusion (Audio + Vibration).
  • Solves acoustic domain shift through a light on-device C++ statistical baseline calibration algorithm.

State of the Art-work

Literature Survey Table

Author's Name / Paper Title	Conference/Journal & Year	Technology / Design	Results Shared by Author	What We Infer (Research Gap)
Lo, Tsai, Yang, Wu → "An Efficient Anomalous Sound Detection System for Microcontrollers"	MDPI Sensors (Nov 2024)	Memory-aware pruned autoencoder for MCU deployment, evaluated on DCASE dataset	Favorable accuracy and resource efficiency (specific AUC not disclosed in abstract)	Audio-only; no vibration fusion or on-device calibration for cross-machine variation
Abbasi, Famouri, Shafiee, Wong → "OutlierNets: Highly Compact Deep Autoencoder Network Architectures for On-Device Acoustic Anomaly Detection"	MDPI Sensors, Jul 2021	Highly compact convolutional autoencoders (as few as 686 parameters, 2.7 KB)	Matches/exceeds larger 4M-parameter models; up to 21–30× lower latency	Audio-only; validated on CPU/ARM Cortex-A72, not a low-cost MCU like ESP32
Chen, Wu, Zhong, Guo, Wu, Ren, Zhang → "DBN-BAAE: Enhanced Lightweight Anomaly Detection Mechanism with Boosting Adversarial Autoencoder"	MDPI Sensors, May 2025	Deep belief network + boosting adversarial autoencoder with dynamic thresholdF1 = 0.82, surpassing best baseline by 1%;	F1 = 0.82, surpassing best baseline by 1%; 2.2× faster training	Targets ICS anomaly detection broadly; no acoustic sensing, no physical MCU hardware validation
Arciniegas, Rivero, Piñan, Diaz, Rivas → "IoT device for detecting abnormal vibrations in motors using TinyML"	Discover Internet of Things, 2025	MPU6050 + ESP32-S3 (XIAO), TinyML via Edge Impulse	25 ms inference on-device; ~300 ms end-to-end alert latency	Vibration-only; no acoustic sensing, no on-device baseline calibration across machines

State of the Art-work

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Kolok, Hodoň, Ševčík, Hotz, Remy → "Low-Cost IoT-Based Predictive Maintenance Using Vibration"	MDPI Sensors (2025)	ESP32 MCU + INMP441 I2S Mic + Accelerometer; RMS & 1024-pt FFT feature extraction	Low-cost (< $20 BOM) hardware feasibility for mechanical imbalance & bearing vibration	Explicitly deferred acoustic processing channel; left joint multi-modal feature fusion & on-device calibration as future work.
Garay, Miranda Bonomi, Mansilla, Fagre, Guzmán, Ritorto, Perez, Katz → "A Multimodal TinyML-Based Predictive Maintenance Architecture for Industrial IoT in the 6G Era"	MDPI Sensors (2026)	Cortex-M4F MCU + Syntiant NDP120 Neural Coprocessor; INT8 Autoencoder (Audio + Vibration + Thermal)	F_1 score = 0.98$ for multi-modal anomaly detection under 6G URLLC	Requires dedicated neural coprocessor ASIC ($100+ hardware) and thermal camera; unoptimized for single low-cost $3 MCUs.
Ūselis et al. Ūselis, Serackis, Pomarnacki → "Signal Processing Optimization in Resource-Limited IoT for Fault Prediction in Rotating Machinery"	MDPI Electronics (2025)	RP2040 MCU + Distilled XGBoost + PCA; Vibration & Motor Current signals	AUC = 0.65–0.78 under domain shift conditions (speed/noise shifts)	Shows severe performance drop (AUC drops 15–25%) when noise/mounting shifts occur; lacks on-device adaptive baseline calibration.
Harada, Niizumi, Takeuchi, Ohishi, Yasuda, Saito → "ToyADMOS2: Another Dataset of Miniature-Machine Operating Sounds for Anomalous Sound Detection under Domain Shift Conditions"	DCASE Workshop (2021)	Unsupervised MobileNetV2 & Autoencoders on acoustic machine operating sounds	25 ms inference on-device; ~300 ms end-to-end alert latency	Audio-only benchmark; recorded with expensive multi-mic arrays; lacks physical vibration fusion and MCU execution.

State of the Art-work

Literature Survey Table

Author's Name / Paper Title	Conference/Journal & Year	Technology / Design	Results Shared by Author	What We Infer (Research Gap)
Purohit, Tanabe, Ichige, Endo, Nikaido, Suefusa, Kawaguchi → "MIMII Dataset: Sound Dataset for Malfunctioning Industrial Machine Investigation and Inspection"	DCASE Workshop (2019)	8-Channel Mic Array + Deep Autoencoder (DAE)	AUC: 0.75–0.95 across 6 dB to -6 dB factory noise	Audio-only benchmark recorded with high-end mic arrays; lacks vibration fusion & MCU execution.
W. A. Smith & R. B. Randall CWRU Benchmark Survey & Evaluation	Elsevier MSSP 10.1016/j.ymssp.2015.04.021	Benchmark vibration analysis of inner/outer race bearing faults	Established benchmark diagnostic procedures	Purely vibration-based; does not fuse acoustic sound to catch early dry-friction squeaks.
J. Chen et al. → Lightweight 1D-CNN for Bearing Fault Diagnosis on Edge Devices	IEEE Trans. on Instrumentation & Measurement (2024) 10.1109/TIM.2024.3361284	Quantized 1D-CNN running directly on ARM Cortex-M / Microcontrollers on CWRU Bearing Dataset	Accuracy: **98.2%**, Model size: **<50 KB**, Latency: **18 ms**	Supervised classification requiring labeled fault datasets; cannot perform unsupervised zero-shot anomaly detection on unseen machines.
H. Zhang et al. → Noise-Robust Vibration Anomaly Detection via Autoencoders	MDPI Sensors (2025) 10.3390/s25020537	Unsupervised Dense/Convolutional Autoencoder trained on healthy CWRU vibration baseline	ROC-AUC: **0.954** on outer & inner race fault detection	Tested only on static bench data; lacks an on-device dynamic calibration mechanism to adapt to speed/load shifts across different factories.

Research Gap

1. **Noise Robustness Gap →** Audio-only detection systems degrade sharply under factory floor noise (85dB+).

- Addressed by: ***fusing acoustic and vibration signals***, so a fault is confirmed across two independent sensing channels rather than one.

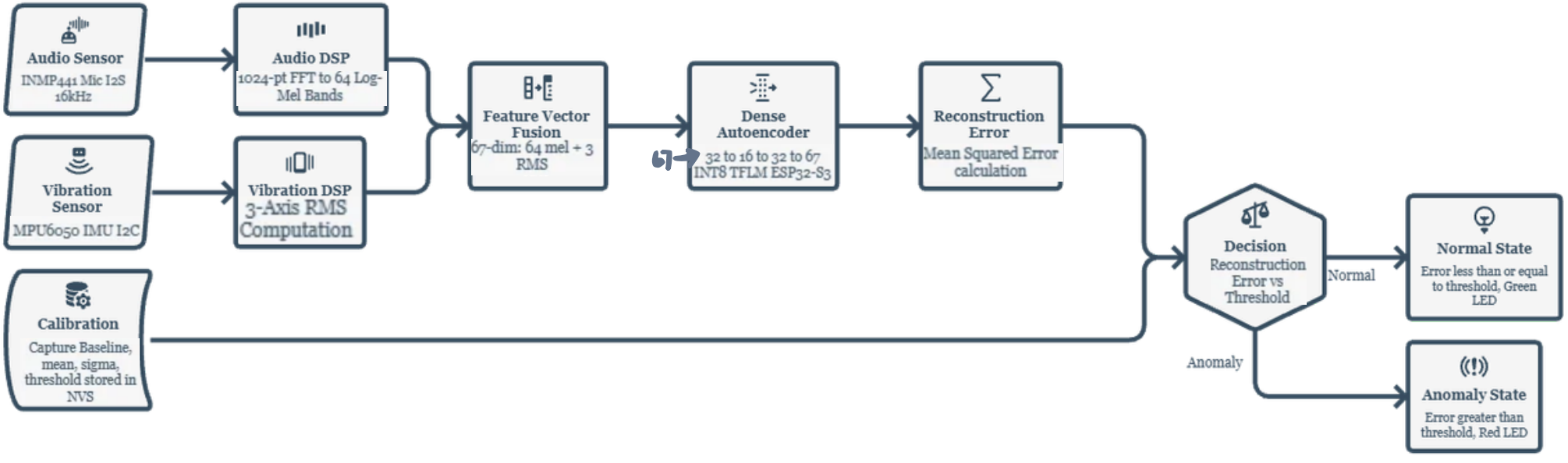
2. **Cross-Machine Generalization Gap →** Fixed, pre-trained models fail when deployed on a different motor than they were trained on.

- Addressed by: an ***on-device baseline calibration routine*** that learns each machine's healthy signature at setup time, rather than assuming one model fits all.

3. **Real-World Validation Gap →** Most published work is evaluated only in simulation or on benchmark audio clips.

- Addressed by: ***physical deployment and testing on an actual motor test rig***, measuring ***real hardware*** performance rather than simulated results.

System Architecture and Workflow



1. Audio (INMP441 mic) and vibration (MPU6050) sensors capture data simultaneously.
2. Signals are converted into a combined feature vector (FFT/log-mel bands + vibration RMS).
3. A lightweight autoencoder reconstructs the expected "healthy" pattern; reconstruction error above a threshold flags an anomaly.
4. Result is shown instantly via an on-device LED (Normal / Anomaly) ,completely offline.

Proposed methodology

→ Model:

- Dense autoencoder, Input(67, Linear) → Encoder Dense(32, ReLU) → Bottleneck Dense(16, ReLU) → Decoder Dense(16, ReLU) → Dense(32, ReLU) → Output Dense(67, Linear). Unsupervised training on healthy-only data; anomaly score = MSE reconstruction loss.

→ DSP Pipeline:

- On-chip 1024-point FFT + 64-band log-mel spectrogram via esp-dsp, fused with 3-axis vibration RMS into a 67-dim feature vector.

→ Inference Engine:

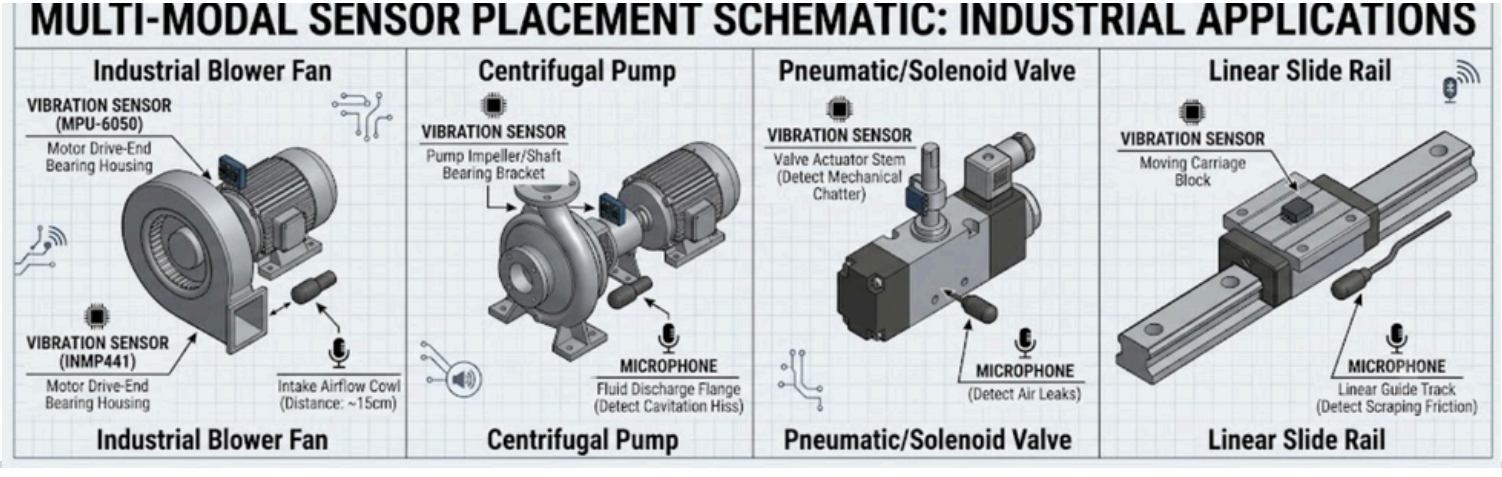
- TensorFlow Lite for Microcontrollers (TFLM), C++ API, running the quantized model directly on-chip.

→ Calibration Logic:

- Statistical baseline calibration (mean μ, std dev σ) computed per-device at setup; dynamic threshold τ = μ_error + 3σ_error, persisted in ESP32 non-volatile storage (NVS).

→ Datasets & Benchmarks:

- Hitachi MIMII Industrial Dataset (Primary Acoustic Benchmark: Fans & Valves with real factory background noise)
- Case Western Reserve University (CWRU) Bearing Dataset (Primary Vibration Benchmark: 3-axis accelerometer baseline & fault signals)



Technology Stack Specification

→ **Hardware Stack Components:**

- Microcontroller: ESP32-S3-N8R8 (Dual-core LX7 @ 240MHz, 512KB SRAM, 8MB Flash, Xtensa Vector DSP Acceleration).
- Acoustic Sensor: INMP441 Digital I2S MEMS Omnidirectional Microphone Module.
- Vibration Sensor: MPU6050 6-Axis Accelerometer & Gyroscope Module (I2C).
- Power Measurement: INA219 Current Sensor / USB Hardware Power Meter.

→ **Machine Learning & Signal Processing Stack (Laptop Training):**

- Programming Language: Python 3.10+
- Frameworks: PyTorch / TensorFlow 2.x / Keras / Librosa / SciPy / NumPy.
- Optimization: TensorFlow Lite Converter (INT8 Post-Training Quantization & QAT).
- Evaluation Metrics: Scikit-Learn (ROC-AUC, Precision, Recall, F1-Score, Confusion Matrix).

→ **Embedded Software Stack (On-Chip Execution):**

- Development Environment: Arduino IDE / ESP-IDF (C++).
- DSP Acceleration: `esp-dsp` C++ library (Hardware Vector FFT & Mel-Filterbank).
- On-Device ML Engine: TensorFlow Lite for Microcontrollers (TFLM C++ API).
- Storage: ESP32 Non-Volatile Storage (NVS / `Preferences.h`).

Implementation plan

Phase 1 → Model Development: Preprocess MIMII audio with Librosa; extract log-mel features; train and validate the dense autoencoder in PyTorch/TensorFlow; evaluate using ROC-AUC, Precision, Recall, F1 on held-out healthy/anomalous samples.

Phase 2 → Model Compression & Deployment: Apply INT8 post-training quantization; convert to TFLite Micro format; integrate into ESP32-S3 firmware via the TFLM C++ API; validate inference correctness against the Python reference model.

Phase 3 → Firmware & DSP Integration: Implement I2S DMA audio capture, esp-dsp FFT/log-mel pipeline, and I2C vibration reading in C++; fuse both streams into the 67-dim feature vector feeding the on-chip model.

Phase 4 → Calibration & Threshold Logic: Implement the baseline-capture routine, μ/σ computation, and dynamic threshold logic; persist calibration state to NVS flash.

Phase 5 → Hardware Assembly & Validation: Integrate sensors, microcontroller, and LED indicators into an enclosure; mount on a physical motor test rig; validate detection performance under real operating conditions and background noise

Innovations, Objectives & Expected Outcomes

→ Objectives

- Design and build a standalone, low-cost (~₹1,500) per Edge AI device for industrial machine anomaly detection.
- Achieve real-time, fully offline inference on the ESP32-S3, no cloud dependency.
- Implement on-device calibration so the same model adapts to different machines without retraining.
- Validate detection performance (targeting >90% ROC-AUC) on a physical motor test rig using the MIMII dataset.

→ Innovation / Novelty

- Fuses acoustic and vibration sensing together, where most existing microcontroller-based work (see Literature Survey) is audio-only and fails under factory noise.
- On-device baseline calibration removes the need for per-machine retraining, a gap explicitly unsolved in the reviewed literature.
- Targets a $3-class microcontroller (ESP32-S3), where comparable multi-modal approaches (e.g. Gupta et al.) require Raspberry Pi-class hardware.
- Validated on physical hardware rather than software simulation alone, addressing a gap common across the reviewed papers.

→ Expected Outcomes

- A working, standalone hardware prototype operating 100% offline.
- Demonstrated noise robustness through sensor fusion, validated on a real motor rig.
- A cost-effective alternative (~₹1,500) per device to commercial systems priced at ₹50,000+.
- A research manuscript targeted for IEEE/Springer conference submission.

Feasibility Evaluation and Project Planning

Technical and Resource Feasibility

Hardware: All core components → ESP32-S3 microcontroller, INMP441 MEMS microphone, and MPU6050 IMU are sourced from established Indian suppliers (Robu.in, Robocraze, Probots) with 2–3 day delivery, eliminating dependency on international procurement. Total hardware cost is aprrox ~1500/- per device, (6k for all four components)
Software: The development stack is entirely open-source, comprising Python and PyTorch for model training, Arduino IDE/ESP-IDF for embedded firmware, and TensorFlow Lite for Microcontrollers for on-device inference requiring no proprietary licensing or cloud infrastructure.
Dataset: The Hitachi MIMII industrial sound dataset is publicly available and directly applicable to the target use case, covering representative machine classes (fans, pumps, valves, slide rails).
Compute Requirements: Given the compact size of the proposed autoencoder, model training is feasible on standard laptop-class hardware, without recourse to GPU clusters or cloud compute.

Timeline & Milestones

This Semester - Design & Documentation Phase

- Month 1 : Problem definition, literature survey, research gap identification is being covered in this Zeroth Review
- Month 2 : System architecture design, DSP pipeline design, autoencoder architecture finalization
- Month 3 : Calibration logic design, feasibility study, BOM finalization, MIMII dataset study
- Month 3.5 : Phase-I report compilation and final Phase-I review preparation

Next Semester - Implementation & Validation Phase

- Month 4 : Dataset preprocessing, offline model training and validation
- Month 5 : Model quantization (INT8), firmware development, DSP integration on ESP32-S3
- Month 6 : Calibration logic implementation, hardware assembly
- Month 6.5–7 : Physical test rig validation, benchmarking, final documentation and manuscript

Feasibility Evaluation and Project Planning

Risk & Mitigation

Risk	Mitigation
• Sensor noise and domain shift across different motors	• On-device calibration adapts the detection threshold per machine, rather than relying on a fixed pre-trained model
• Limited access to industrial motors for physical validation	• Phased validation strategy : initial testing on the MIMII dataset, followed by evaluation on an available lab motor test rig
• Integration delays from combining DSP and inference on a single chip	• Reliance on pre-validated libraries (esp-dsp, TFLM) in place of custom signal-processing implementation

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